Beyond the Leaderboard: What Makes Deer Demographics Recognizable in Snapshot Wisconsin Trail-Camera Images?

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1. Problem definition

We will predict two labels for white-tailed deer in trail-camera images: *antler presence* (antlered/antlerless) and *age group* (adult/young). The Snapshot Wisconsin "Oh Deer" challenge provides 4,880 labeled images and animal bounding boxes, with F1-based leaderboard evaluation [**?** **?**]. According to the challenge briefing, the available boxes originate from a MegaDetector detection pipeline [**?**]. We ask: *How do pretraining, animal cropping, and seasonal context affect these predictions, particularly at camera sites unseen during training?* We will treat antler presence as an observed visual attribute, not as a direct substitute for sex.

2. Motivation

Snapshot Wisconsin has collected more than 100 million images, creating a practical need for scalable wildlife annotation [**?**]. Deer age-related observations are relevant to population monitoring, but visual categorization is difficult: fawns change appearance as they mature; small or occluded antlers can be ambiguous; and night-time infrared imagery limits visible detail. Even human observers make systematic age/sex classification errors on white-tailed deer photographs [**?**]. Models can also exploit fixed camera backgrounds and lose accuracy at novel locations [**?**]. Prior work demonstrates the feasibility of learning age and sex labels for *red deer* in a different environment [**?**] and the value of detector-based crops in camera-trap classification [**?**]. Our contribution is a controlled study of which signals improve *white-tailed deer* age and antler prediction without encouraging location-specific shortcuts.

3. Tentative approach

We will use a shared image encoder with two binary classification heads. Three focused comparisons will guide the experiments:
(1) Pretrained features. Compare an ImageNet-pretrained ResNet-50 baseline with DINOv2 [**?**] and biologically pretrained BioCLIP 2 [**?**]. We will evaluate frozen-feature linear probes under matched data splits and selectively fine-tune promising backbones. For BioCLIP 2, we will also test natural-language zero-shot prompts, a capability of CLIP-style vision–language models [**?**].
(2) Localization. Compare whole-frame inputs with crops from the *provided detector-generated boxes*, including a padded-crop variant that retains nearby context. This avoids treating detector predictions as human-verified ground truth. We will inspect misclassifications and, if time permits, attention maps to assess whether backgrounds dominate the decision.
(3) Seasonal context. Compare image-only predictions with late fusion of month/season, where capture dates are available. We will test performance both with and without metadata at inference, and will *not* use raw camera-site IDs as predictors for unseen-site evaluation.

4. Evaluation plan

Metrics. Report the challenge F1 score when its exact evaluation implementation is available; independently report macro-F1 for each label, class-wise precision/recall, confusion matrices, and inference latency. **Splits.** Compare a within-site split with a camera-site-held-out split, ensuring images from the same capture burst stay together wherever burst identifiers or timestamps allow. The latter protocol requires usable site metadata; if these are unavailable, we will report that limitation rather than claim cross-site generalization. **Ablations.** Hold preprocessing and training budgets fixed while varying backbone, crop type, and metadata. We will check performance by day/night and season, where those labels are available. **Expected outcome.** Identify whether biological pretraining, localized evidence, or temporal context offers gains that persist at unseen cameras. As a stretch goal, we will test one additional ecologically useful attribute (e.g., visible snow), provided labels can be obtained; this also addresses the challenge's usefulness criterion. Baselines will be completed first, followed by location and metadata ablations for the final report.